

Introduction to the Ghana Breast Health Study

1.1 Background

Breast cancer is becoming more common in many African countries, but there have been few comprehensive studies of the disease in African populations. The Ghana Breast Health Study is an epidemiologic investigation of potential risk factors for breast cancer in Ghanaian women. The study will be conducted in the capital, Accra, in collaboration with investigators from Korle Bu Teaching Hospital (KBTH) and the University of Ghana Medical School, and in the second largest city, Kumasi, in collaboration with investigators from Komfo Anokye Teaching Hospital (KATH) and Peace and Love Hospital (PLH). Study participants are asked to provide detailed lifestyle and health history information using an interviewer-based questionnaire. Additionally, anthropometry measurements are taken and participants are asked to provide blood, saliva, and stool samples for assessment of biologic markers. Tumor tissue samples are sought from cases in order to classify them into subtypes.

We will compare data obtained from women with incident breast cancer (first breast cancer diagnosis within the past year) to data from women with no history of breast cancer in order to learn more about what factors are associated with breast cancer in Ghana. The study is described in the study brochure (Appendix 1-1).

1.2 Organizational Structure

1.2.1 National Institutes of Health (NIH), National Cancer Institute (NCI)

The study is being conducted by the Hormonal and Reproductive Branch, Division of Cancer Epidemiology and Genetics of the National Cancer Institute (NCI), which is part of the National Institutes of Health (NIH). The NCI Investigators, Drs. Louise Brinton and Jonine Figueroa, are Co-Principal Investigators responsible for design and oversight of the study. They provide support for the development and implementation of the study protocol.

1.2.2 Ghanaian Hospital Study Sites

The study is being conducted at three major hospitals in Ghana as follows:

1. Korle Bu Teaching Hospital in Accra led by Dr. Joe-Nat Clegg-Lamptey
2. Komfo Anokye Teaching Hospital in Kumasi led by Dr. Baffour Awuah
3. Peace and Love Hospital in Kumasi led by Dr. Beatrice Wiafe Addai

1.2.3 Westat

Westat works closely with NCI and the three hospitals in Ghana to ensure proper data collection and protocol adherence in order to ensure the overall success of the study. Westat assisted with the development of the study and produced the Interviewer Manual and Manual of Operations and Procedures. Westat trained six Interviewers and two Study Managers from each site during an intensive 4-day session that took place in Kumasi, Ghana in January 2013 prior to the start of the study.

Westat supports NCI and the three hospital sites by monitoring study recruitment and completion of study tasks, communicating revisions to the study protocol, and tracking study decisions. Westat maintains regular email and telephone contact with the Study Managers at each hospital. Westat monitors the inventory of study supplies at each hospital, purchases additional supplies as needed, and ships supplies to the sites.

Westat updates the data entry system as needed, manages raw data, and creates analytic datasets. Westat also receives and compiles scanned images of data collection forms. Westat monitors quality metrics within the data and communicates with the Study Managers to request information needed for data cleaning. Westat also generates specimen manifests for shipment of specimens to the NCI biorepository and reports on the specimen inventory.

1.3 Study Design

This study will use a case-control design. Cases will be recruited from Korle-Bu Teaching Hospital in Accra, and Komfo Anokye Teaching Hospital and Peace and Love Hospital, both in Kumasi. It is

estimated that the vast majority of diagnosed breast cancer patients in Ghana who seek treatment will be seen at these three hospitals, as they are the primary facilities in the country offering specialized breast cancer treatment.

Population controls will be frequency-matched to cases based on the distribution of expected age and district of residence. For the pilot feasibility phase of the study, which took place between February and September 2013, controls were sampled in a case:control ratio of 1:2. Based on review by NCI's Scientific Advisory Group in July 2013, the sampling for controls was reduced to a 1:1 ratio for the entire study overall.

Based on the lessons learned during the feasibility phase of the study, a review was undertaken of study procedures and forms in October-December 2013. The following changes were made and circulated:

- The Anthropometry calibration protocol was revised and circulated in September 2013.
- Minor revisions to the Blood Processing protocol affecting storage of samples were distributed on November 1, 2013.
- The Stool Collection and Processing protocol was not initiated until November 2013 at which time a new protocol V1.1 was instituted.
- Version 2.0 of the Eligibility Screener Form, Recruitment Outcome Form, Risk Factor Questionnaire, and Breast Biopsy Form were distributed for use beginning on January 13, 2014.
- The Twi version of the Risk Factor Questionnaire ceased to be used in December 2013.
- A revised Stool Collection and Processed protocol Version 2.0 allowing for stool collection for controls in the field was distributed on February 4, 2015.

1.4 Study Components

For each participant that agrees to be in the study and provides Informed Consent for all components, the following is a list of approximate times to complete each task. Due to protocol issues, the stool was not collected as part of the feasibility phase of the study but was initiated at all sites in November–December 2013.

■ Eligibility Screener	15 minutes
■ Recruitment Form	10 minutes

■ Risk Factor Questionnaire	60 minutes
■ Anthropometry	20 minutes
■ Saliva Collection	10 minutes
■ Stool Collection Explanation	10 minutes
■ Stool Collection	10 minutes
■ Stool Processing	5 minutes
■ Blood Collection	15 minutes
■ Blood Processing	20 minutes
■ Breast Tissue Biopsy (Cases Only)	N/A
■ Breast Tissue Processing (Cases Only)	N/A
■ Breast Case Abstract Form (Cases Only)	30 minutes

1.5 Biospecimen Management

From initiation of the feasibility study in February 2013 until June 5, 2014, blood samples were processed and stored in -80°C freezers at KBTH and KATH only. Samples collected at Peace and Love Hospital were transferred to KATH for processing and storage prior to shipment to the NCI biorepository in Frederick, MD. Upon installation of a freezer at PLH on June 6, 2014, blood samples are processed and stored in -80°C freezers at each site. Similarly, stool samples collected at PLH from stool protocol initiation on November 14, 2013 until June 5, 2014 were transferred to KATH for processing and storage. From June 6, 2014 forward, PLH stool samples are processed and stored at PLH. Saliva samples are stored at ambient temperature at each site from collection until shipment to NCI.

The assays for the study will be conducted in the U.S. Only the blood clot aliquot will remain at the study hospitals to enable researchers to perform assays and increase capacity for molecular research at their labs in the future. The samples shipped to the U.S. are shipped on a regular schedule to the NCI repository or to the selected laboratories for analysis. Refer to Chapter 12 for details about shipping.

1.6 Data Management

A data entry system was developed by Westat for entry of all information collected on study forms and the questionnaire. Data entry staff from each site were trained in January 2013 on how to use the system to enter data and scan images of the forms. A Westat data manager and systems staff extract the data at scheduled intervals, make updates to the system as needed, review the entered data for QC purposes, and prepare analysis datasets. Westat data management staff also download the scanned form images on a weekly basis and periodically review images to resolve data issues.

1.7 Timeline

Training of study staff in Ghana took place in January 2013. The feasibility phase of the study began in February 2013 and was completed in September 2013. Evaluations were made along the way to troubleshoot issues that arose and to determine the best way to proceed with the study. The study was approved by the NCI Scientific Advisory Group for two additional years following the feasibility phase. The main phase of the study began recruitment at each site starting in September 2013 and completion is anticipated between September 2015 and December 2015. There was no break in enrollment or formal change in protocol to denote the shift from the feasibility phase to the main phase of the study.